XSS

CSE 405 January 2025 - Lecture 11

Courtesy of: https://textbook.cs161.org/

Last Time: Cookies

- Cookie: a piece of data used to maintain state across multiple requests
  - Set by the browser or server
  - Stored by the browser
  - Attributes: Name, value, domain, path, secure, HttpOnly, expires
- Cookie policy
  - Server with domain X can set a cookie with domain attribute Y if the domain attribute is a **domain suffix** of the server's domain, and the domain attribute Y is not a top-level domain (TLD)
  - The browser attaches a cookie on a request if the domain attribute is a **domain suffix** of the server's domain, and the path attribute is a **prefix** of the server's path
  - Cookie domain: example.com and cookie path: /some/path will be included in request to https://foo/example/com/some/path/index.html

Last Time: Session Authentication

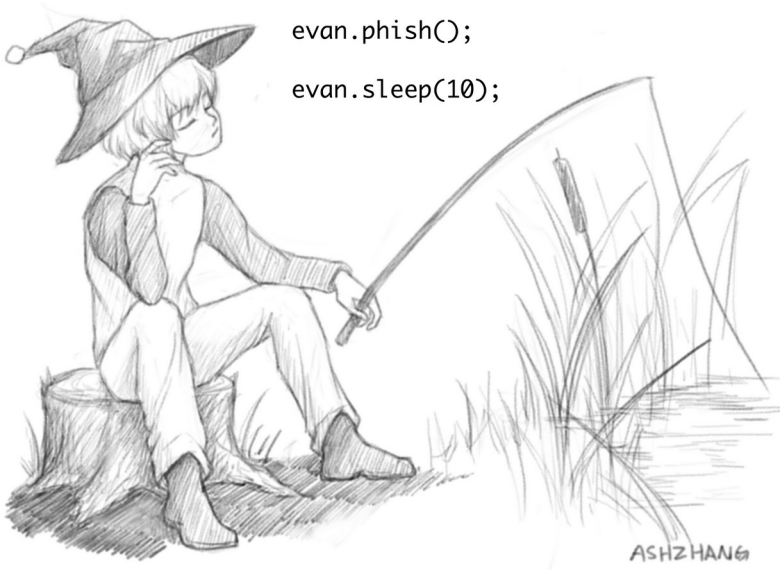
- Session authentication
  - Use cookies to associate requests with an authenticated user
  - First request: Enter username and password, receive session token (as a cookie)
  - Future requests: Browser automatically attaches the session token cookie
- Session tokens
  - If an attacker steals your session token, they can log in as you
  - Should be randomly and securely generated by the server
  - The browser should not send tokens to the wrong place

Last Time: CSRF

- Cross-site request forgery (CSRF or XSRF): An attack that exploits cookie-based authentication to perform an action as the victim
  - User authenticates to the server
    - User receives a cookie with a valid session token
  - Attacker tricks the victim into making a malicious request to the server
  - The server accepts the malicious request from the victim
    - Recall: The cookie is automatically attached in the request
- Attacker must trick the victim into creating a request
  - GET request: click on a link
  - POST request: use JavaScript

Today: XSS

- XSS
  - Websites use untrusted content as control data
  - Stored XSS
  - Reflected XSS
  - Defense: HTML sanitization
  - Defense: Content Security Policy (CSP)
- UI attacks
  - Clickjacking
  - Phishing



Cross-Site Scripting (XSS)

Textbook Chapter 22

Top 25 Most Dangerous Software Weaknesses (2020)

Rank	ID	Name	Score
[1]	CWE-79	Improper Neutralization of Input During Web Page Generation ('Cross-site Scripting')	46.82
[2]	CWE-787	Out-of-bounds Write	46.17
[3]	CWE-20	Improper Input Validation	33.47
[4]	CWE-125	Out-of-bounds Read	26.50
[5]	CWE-119	Improper Restriction of Operations within the Bounds of a Memory Buffer	23.73
[6]	CWE-89	Improper Neutralization of Special Elements used in an SQL Command ('SQL Injection')	20.69
[7]	CWE-200	Exposure of Sensitive Information to an Unauthorized Actor	19.16
[8]	CWE-416	Use After Free	18.87
[9]	CWE-352	Cross-Site Request Forgery (CSRF)	17.29
[10]	CWE-78	Improper Neutralization of Special Elements used in an OS Command ('OS Command Injection')	16.44
[11]	CWE-190	Integer Overflow or Wraparound	15.81
[12]	CWE-22	Improper Limitation of a Pathname to a Restricted Directory ('Path Traversal')	13.67
[13]	CWE-476	NULL Pointer Dereference	8.35
[14]	CWE-287	Improper Authentication	8.17
[15]	CWE-434	Unrestricted Upload of File with Dangerous Type	7.38
[16]	CWE-732	Incorrect Permission Assignment for Critical Resource	6.95
[17]	CWE-94	Improper Control of Generation of Code ('Code Injection')	6.53

Review: Same-Origin Policy

- Two webpages with different origins should not be able to access each other's resources
  - Example: JavaScript on **`http://evil.com`** cannot access the information on **`http://bank.com`**

Review: JavaScript

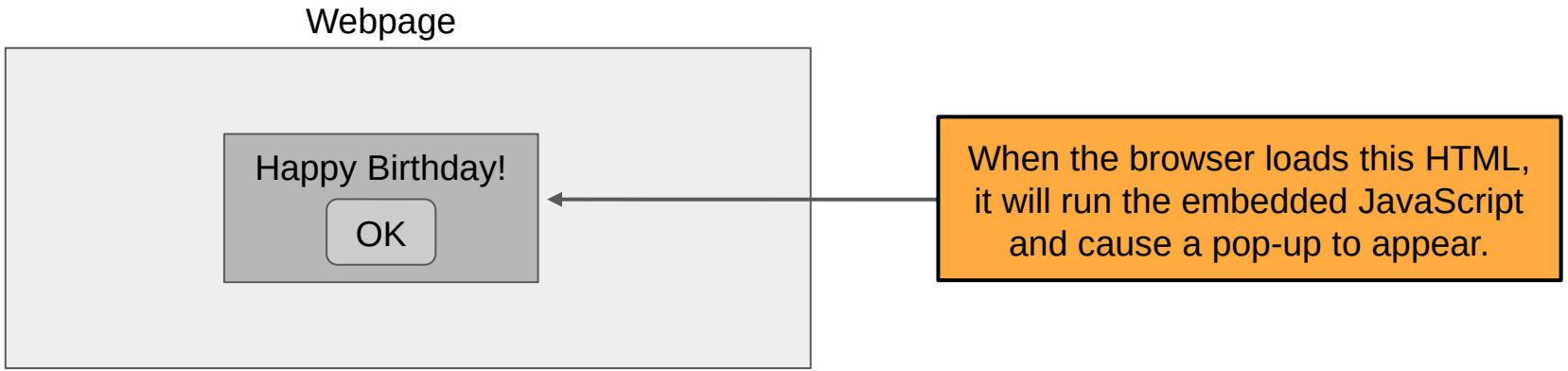
- **JavaScript**: A programming language for running code in the web browser
- JavaScript is **client-side**
  - Code sent by the server as part of the response
  - Runs in the browser, not the web server!
- Used to manipulate web pages (HTML and CSS)
  - Makes modern websites interactive
  - JavaScript can be directly embedded in HTML with **`<script>`** tags
- Most modern webpages involve JavaScript
  - JavaScript is supported by all modern web browsers
- You don’t need to know JavaScript syntax
  - However, knowing common attack functions helps

Review: JavaScript

- JavaScript can create a pop-up message

HTML (with embedded JavaScript)

```
<script>alert("Happy Birthday!")</script>
```



A Go HTTP Handler

Handler

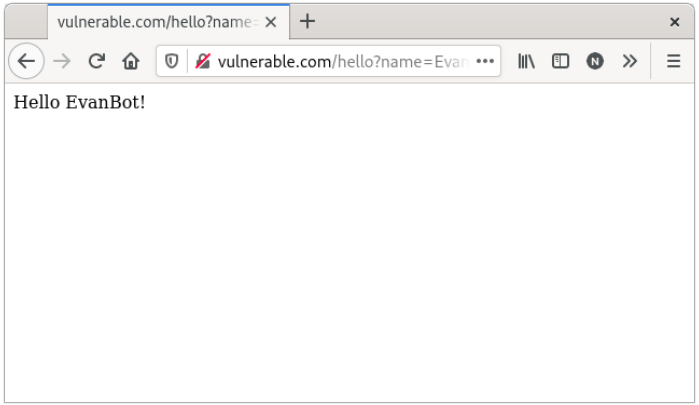
```
func handleSayHello(w http.ResponseWriter, r *http.Request) {
    name := r.URL.Query()["name"][0]
    fmt.Fprintf(w, "<html><body>Hello %s!</body></html>", name)
}
```

URL

```
https://vulnerable.com/hello?name=EvanBot
```

Response

```
<html><body>Hello EvanBot!</body></html>
```



A Go HTTP Handler

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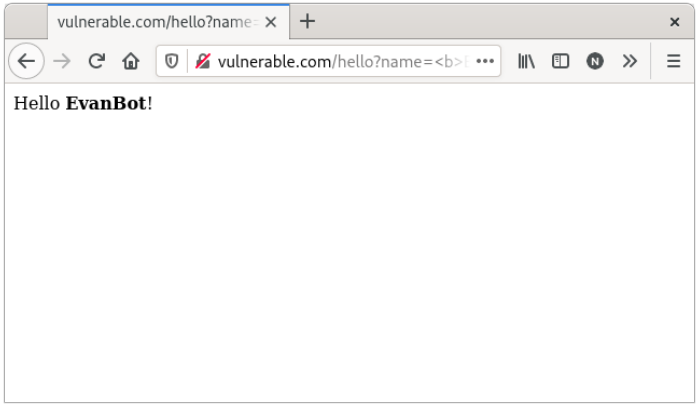
func handleSayHello(w http.ResponseWriter, r *http.Request) {
    name := r.URL.Query()["name"][0]
    fmt.Fprintf(w, "<html><body>Hello %s!</body></html>", name)
}
```

URL

```
https://vulnerable.com/hello?name=<b>EvanBot</b>
```

Response

```
<html><body>Hello <b>EvanBot</b>!</body></html>
```



A Go HTTP Handler

Problem: This input represents control data (HTML), not just text!

Handler

```
func handleSayHello(w http.ResponseWriter, r *http.Request) {
    name := r.URL.Query()["name"][0]
    fmt.Fprintf(w, "<html><body>Hello %s!</body></html>", name)
}
```

URL

```
https://vulnerable.com/hello?name=<script>alert(1)</script>
```

Response

```
<html><body>Hello <script>alert(1)</script>!</body></html>
```

A Go HTTP Handler

Not just **%s**: It can happen with any string manipulation

Handler

```
func handleSayHello(w http.ResponseWriter, r *http.Request) {
    name := r.URL.Query()["name"][0]
    content := "<html><body>Hello "+name+"!</body></html>"
    fmt.Fprint(w, content)
}
```

URL

```
https://vulnerable.com/hello?name=<script>alert(1)</script>
```

Response

```
<html><body>Hello <script>alert(1)</script>!</body></html>
```

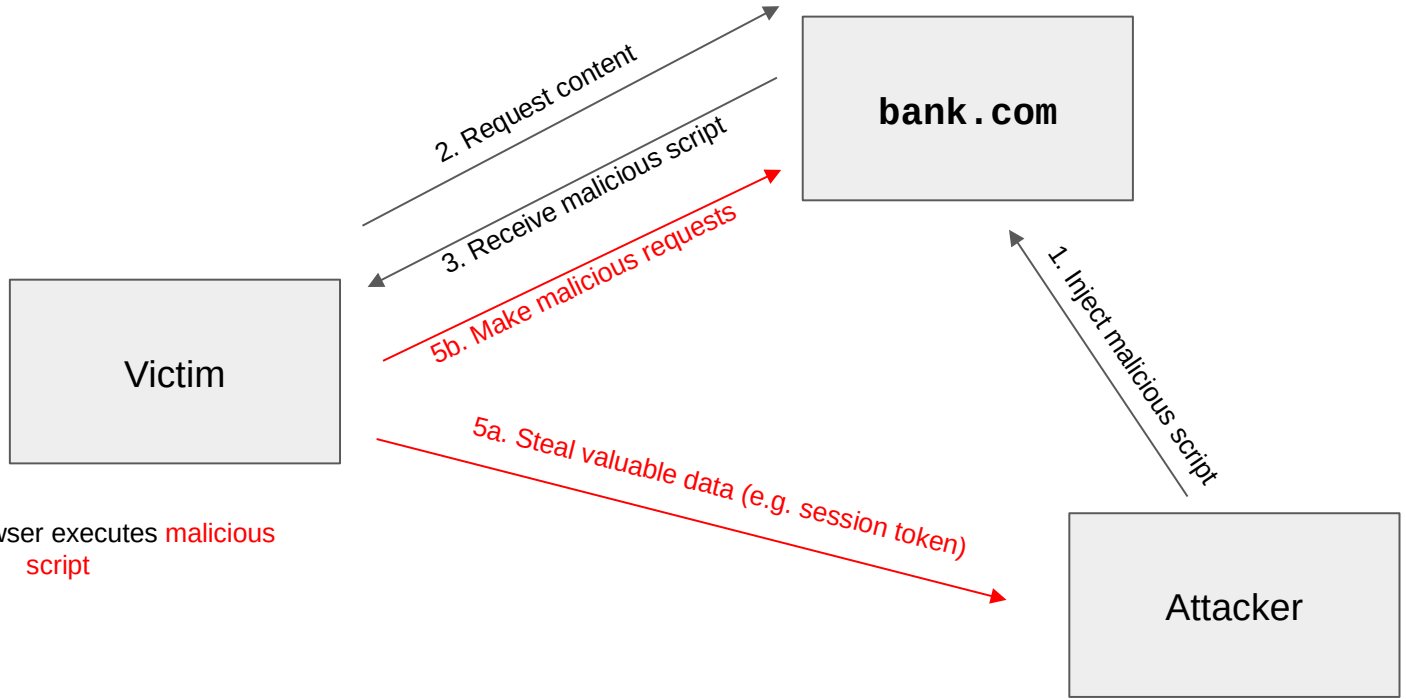
Cross-Site Scripting (XSS)

- Idea: The attacker adds malicious JavaScript to a legitimate website
  - The legitimate website will send the attacker's JavaScript to browsers
  - The attacker's JavaScript will run with the origin of the legitimate website
  - Now the attacker's JavaScript can access information on the legitimate website!
- **Cross-site scripting** (**XSS**): Injecting JavaScript into websites that are viewed by other users
  - Cross-site scripting subverts the same-origin policy
- Two main types of XSS
  - Stored XSS
  - Reflected XSS

Stored XSS

- **Stored XSS** (**persistent XSS**): The attacker's JavaScript is **stored** on the legitimate server and sent to browsers
- Classic example: Facebook pages
  - An attacker puts some JavaScript on their Facebook page
  - Anybody who loads the attacker's page will see JavaScript (with the origin of Facebook)
- Remember: Stored XSS is a server-side vulnerability!

Stored XSS

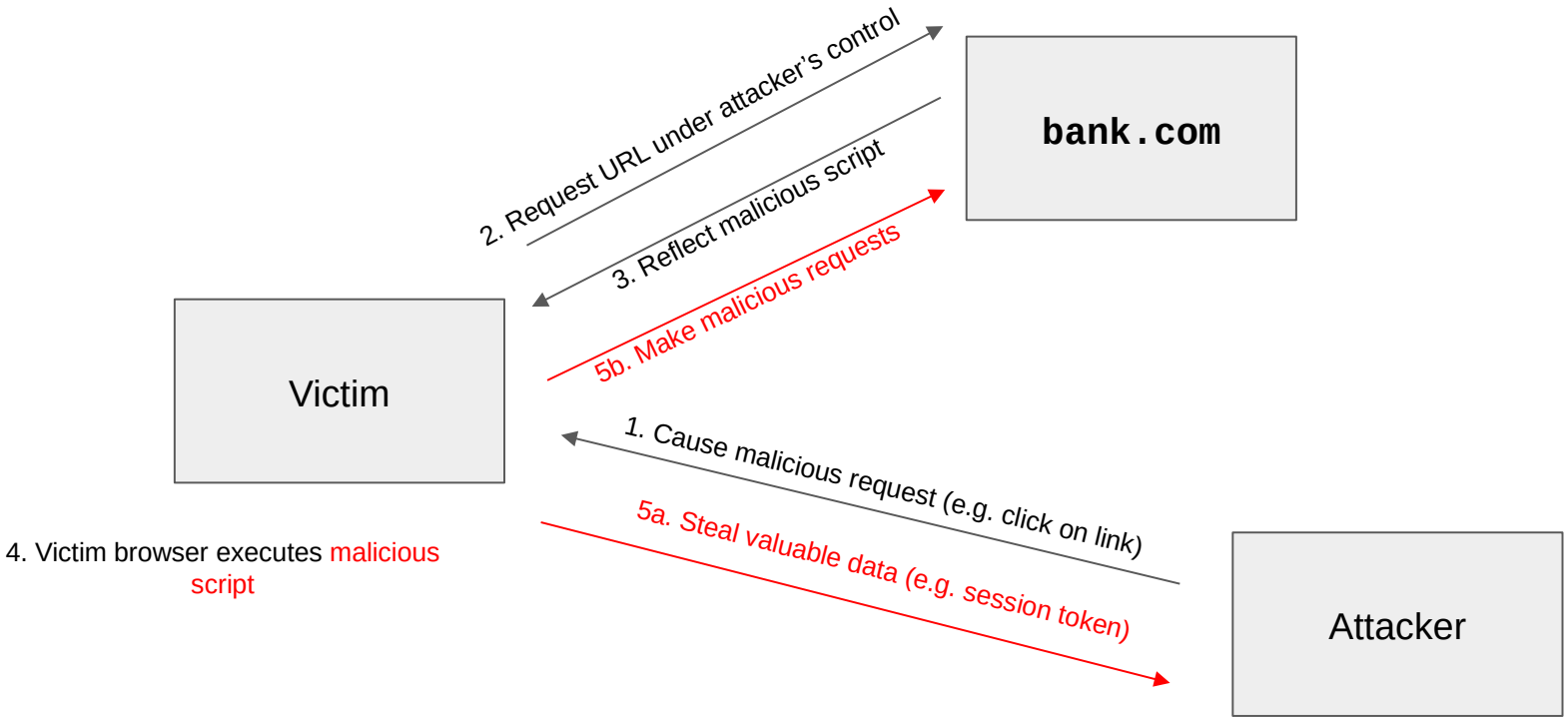


4. Victim browser executes malicious script

Reflected XSS

- **Reflected XSS**: The attacker causes the victim to input JavaScript into a request, and the content is **reflected** (copied) in the response from the server
- Classic example: Search
  - If you make a request to **`http://google.com/search?q=evanbot`**, the response will say "10,000 results for evanbot"
  - If you make a request to **`http://google.com/search?q=<script>alert(1)</script>`**, the response will say "10,000 results for **`<script>alert(1)</script>`**"
- Reflected XSS requires the victim to make a request with injected JavaScript

Reflected XSS



Reflected XSS: Making a Request

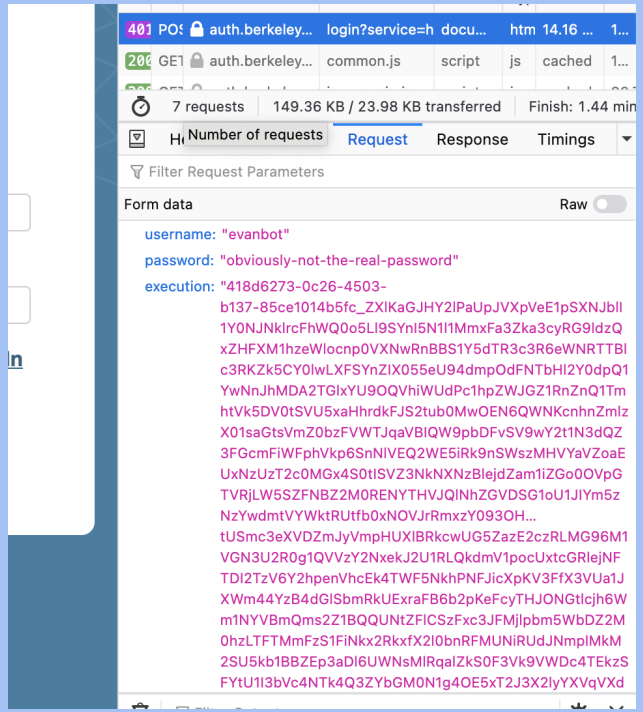
- How do we force the victim to make a request to the legitimate website with injected JavaScript?
  - Trick the victim into visiting the attacker's website, and include an embedded iframe that makes the request
    - Can make the iframe very small (1 pixel x 1 pixel), so the victim doesn't notice it:
      ```
      <iframe height=1 width=1 src="http://google.com/search?q=<script>alert(1)</script>">
      ```
  - Trick the victim into clicking a link (e.g. posting on social media, sending a text, etc.)
  - Trick the victim into visiting the attacker's website, which redirects to the reflected XSS link
  - … and many more!

Reflected XSS is not CSRF

- Reflected XSS and CSRF both require the victim to make a request to a link
  - XSS: An HTTP response contains maliciously inserted JavaScript, executed on the client side
  - CSRF: A malicious HTTP request is made (containing the user's cookies), executing an effect on the server side

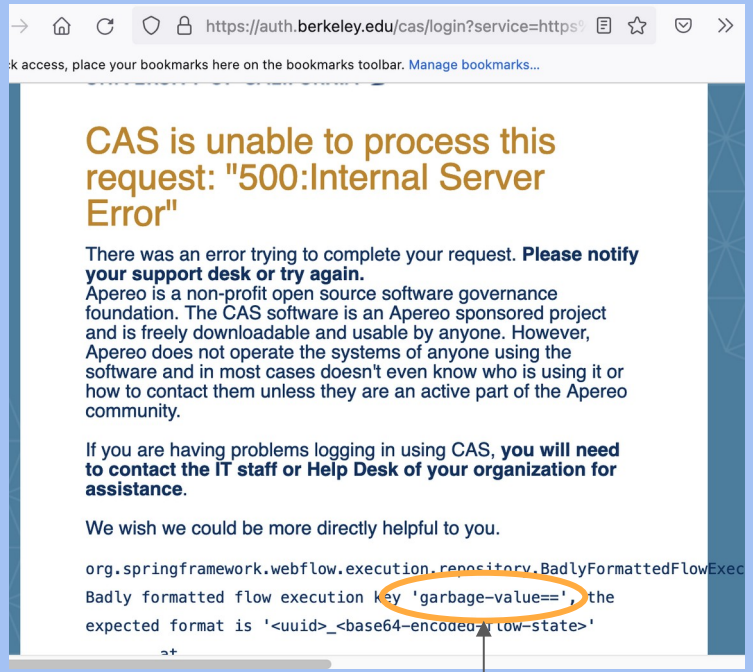
XSS in the Wild… On CalNet?!

- In 2021, 61C student Rohan Mathur was exploring the CalNet login page
- Fields submitted when logging in:
  - **`username`**: What user am I logging in as?
  - **`password`**: What's the user's password?
  - **`execution`**: Server-side state (like a CSRF token with extra data)



XSS in the Wild… On CalNet?!

- The server expects **`execution`** to be in a specific format that contains its server-side state
  - What if you muck with **`execution`**?
- The “corrupted” execution key is placed into the error message in the HTML to aid debugging
  - … and CalNet at the time did not escape the execution key



Constructing an Attack on CalNet

Attack: Force a POST request to CalNet!

```
<html>
  <head>
    <script>
            // When the malicious page finishes loading, automatically submit the form!
      document.addEventListener('DOMContentLoaded', () => {
        document.getElementById('form').submit();
      });
    </script>
  </head>
  <body>
    <!-- Malicious form containing our malicious execution data. -->
    <form id="form" action="https://auth.berkeley.edu/cas/login" method="POST">
      <input name="username" type="text" value="evanbot" />
      <input name="password" type="text" value="obviously-not-the-real-password" />
      <input name="execution" type="text" value="<script>alert('XSS!')</script>" />
    </form>
  </body>
</html>
```

So What Happened?

- CalNet also accepts **`execution`** parameters over URL query parameters
  - A link like **`https://auth.berkeley.edu/cas/login?execution=<script>alert('XSS!')</script>`** would result in the same attack
- It would only fire if you clicked the button to show the error
- … But if clicked, the injected JavaScript could
  - Present a fake login prompt
  - Steal your CalNet password
  - Log you in as the attacker's account
  - Steal your authentication session token
- Root cause: A >5 year old sample page modified by CalNet
  - CalNet uses software from Apereo, which comes with sample pages for logging in
  - Apereo fixed that bug many years before, but sample pages don't get included in bug fixes!

XSS Defenses

- Defense: **HTML sanitization**
  - Idea: Certain characters are special, so server should remove those characters or encode them in a way that will render them harmless
- Escaping: e.g., replace **<** with **<** with **"** with **"**
- Rely on trusted libraries to do this for you

```
<html>
<body>
Hello &lt;script&gt;alert(1)&lt;/script&gt;!
</body>
</html>
```

XSS Defenses: Escaping

Handler

```
func handleSayHello(w http.ResponseWriter, r *http.Request) {
    name := r.URL.Query()["name"][0]
    fmt.Fprintf(w, "<html><body>Hello %s!</body></html>", html.EscapeString(name))
}
```

URL

```
https://vulnerable.com/hello?name=<script>alert(1)</script>
```

Response

```
<html><body>Hello &lt;script&gt;alert(1)&lt;/script&gt;!</body></html>
```

XSS Defenses: Escaping

- If a programmer has to take an extra action for every usage…
  - They are eventually going to forget (e.g., CalNet)
  - Consider human factors!
- Nowadays, escaping is generally achieved through **templating**
  - HTML templates are essentially their own language, where you declare what data goes where
  - The templating engine handles all the escaping internally, automatically

```
<html>
<body>
Hello {{.name}}!
</body>
</html>
```

Example: Golang HTML template

XSS Defenses: CSP

- Defense: **Content Security Policy** (**CSP**)
  - Idea: Instruct the browser to only use resources loaded from specific places
  - Uses additional headers to specify the policy
- Standard approach:
  - Disallow all inline scripts (JavaScript code directly in HTML), which prevents inline XSS
    - Example: Disallow **`<script>alert(1)</script>`**
  - Only allow scripts from specified domains, which prevents XSS from linking to external scripts
    - Example: Disallow **`<script src="https://cs161.org/hack.js">`**
- Also works with other content (e.g. iframes, images, etc.)
- Relies on the browser to enforce security, so more of a mitigation for defense-in-depth

UI Attacks

Textbook Chapter 23

User Interface (UI) Attacks

- General theme: The attacker tricks the victim into thinking they are taking an **intended** action, when they are actually taking a **malicious** action
  - Takes advantage of **user interfaces**: The trusted path between the user and the computer
    - Browser disallows the website itself to interact across origins (same-origin policy), but trusts the user to do whatever they want
  - Remember: Consider human factors!
- Two main types of UI attacks
  - Clickjacking: Trick the victim into clicking on something from the attacker
  - Phishing: Trick the victim into sending the attacker personal information

Clickjacking

- **Clickjacking**: Trick the victim into clicking on something from the attacker
- Main vulnerability: the browser trusts the user's clicks
  - When the user clicks on something, the browser assumes the user intended to click there
- Why steal clicks?
  - Download a malicious program
  - Like a Facebook page/YouTube video
  - Delete an online account
- Why steal keystrokes?
  - Steal passwords
  - Steal credit card numbers
  - Steal personal info

Clickjacking: Download buttons

- Which is the real download button?
- What if the user clicks the wrong one?

Clickjacking

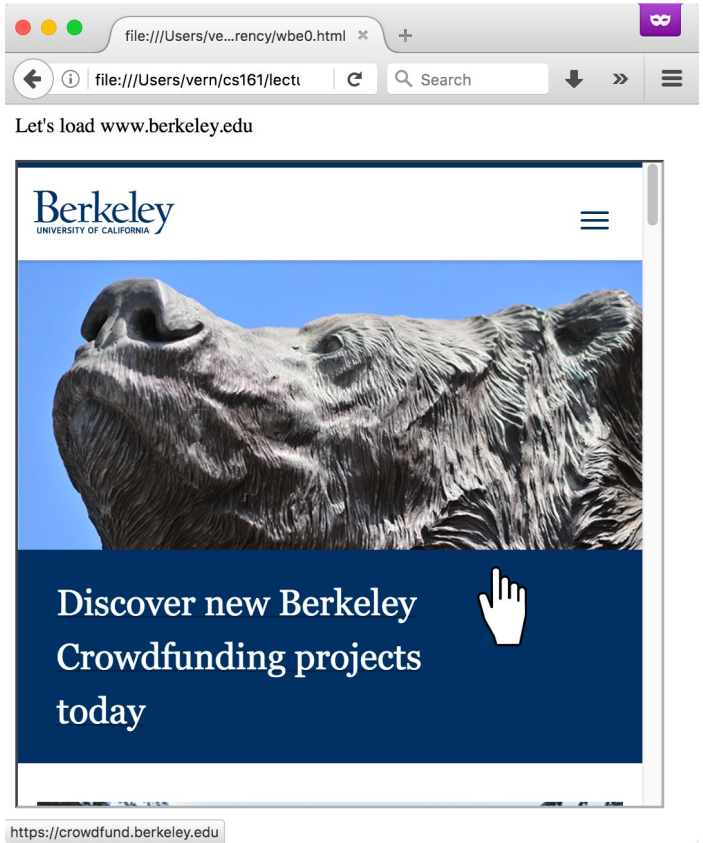
Navigate to berkeley.edu. Notice the URL when hovering over the image.

Clickjacking

Load berkeley.edu in an iframe

We can't generate clicks ourselves because of SOP, but the user can still click…

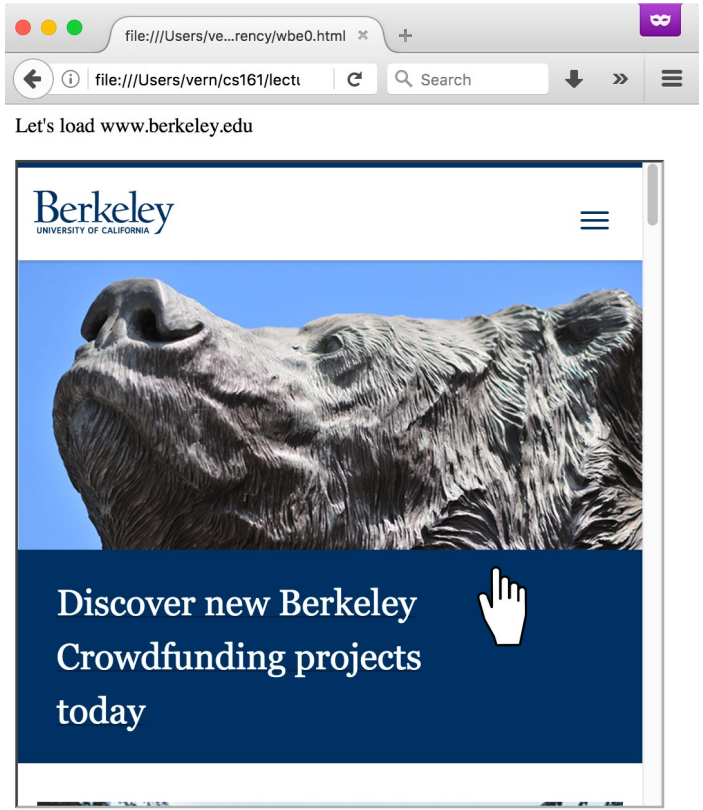
```
<iframe style="opacity: 1.0"
src="https://www.berkeley.edu/"></iframe>
```



Clickjacking

Place some enticing content underneath

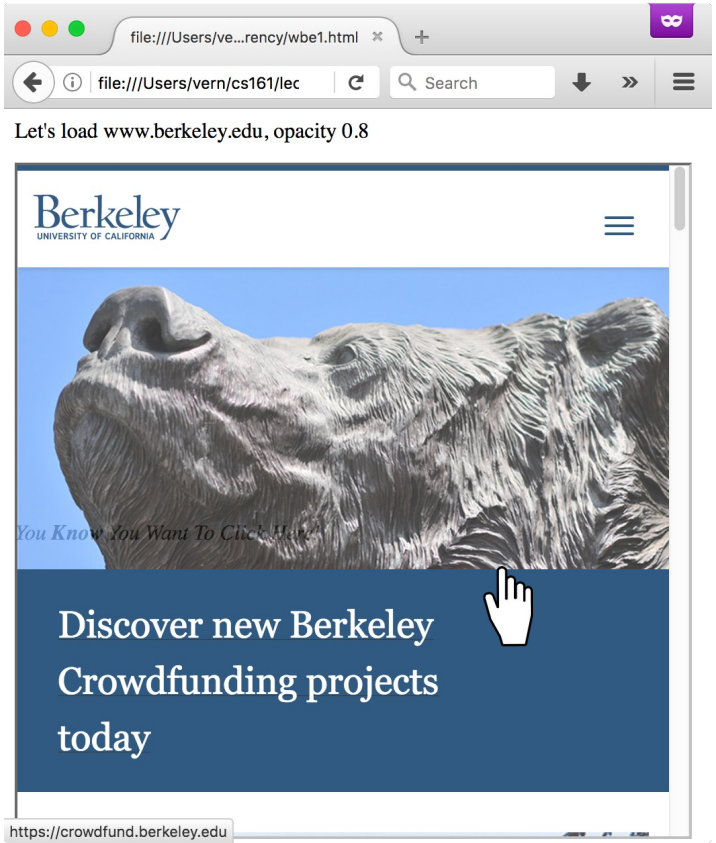
```
<iframe style="opacity: 1.0"
src="https://www.berkeley.edu/"></iframe>
<p style="margin-top: 210pt"><em>You <b>Know</b>
You Want To Click Here!</em></p>
```



Clickjacking

Make the iframe slightly transparent…

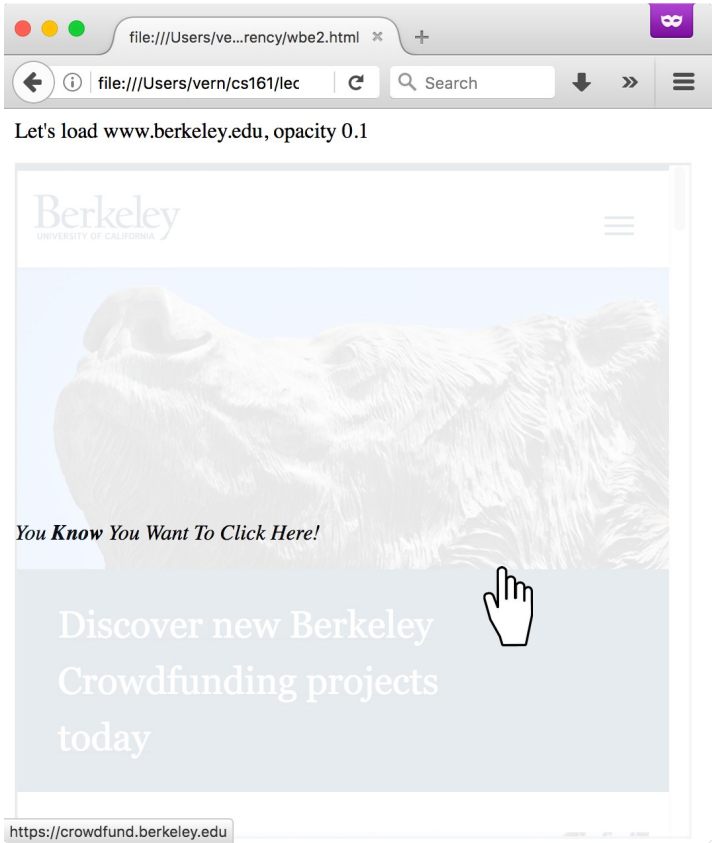
```
<iframe style="opacity: 0.8"
src="https://www.berkeley.edu/"></iframe>
<p style="margin-top: 210pt"><em>You <b>Know</b>
You Want To Click Here!</em></p>
```



Clickjacking

Make it *more* transparent

```
<iframe style="opacity: 0.1"
src="https://www.berkeley.edu/"></iframe>
<p style="margin-top: 210pt"><em>You <b>Know</b>
You Want To Click Here!</em></p>
```

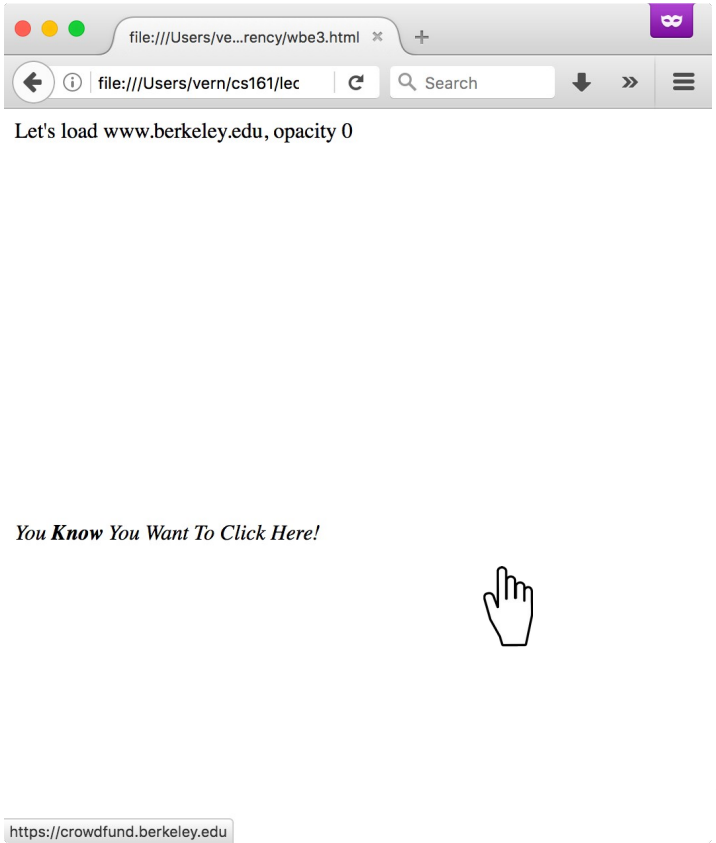


Clickjacking

Make it *entirely* transparent

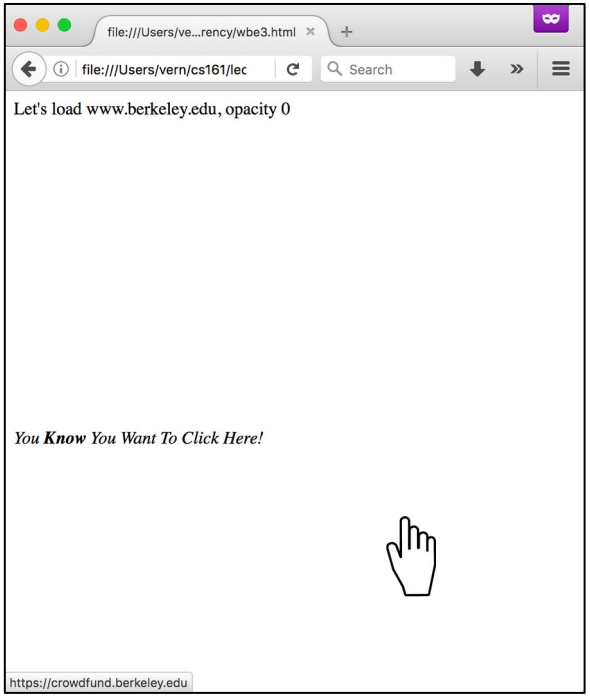
But the user still clicks on the iframe!

```
<iframe style="opacity: 0"
src="https://www.berkeley.edu/"></iframe>
<p style="margin-top: 210pt"><em>You <b>Know</b>
You Want To Click Here!</em></p>
```



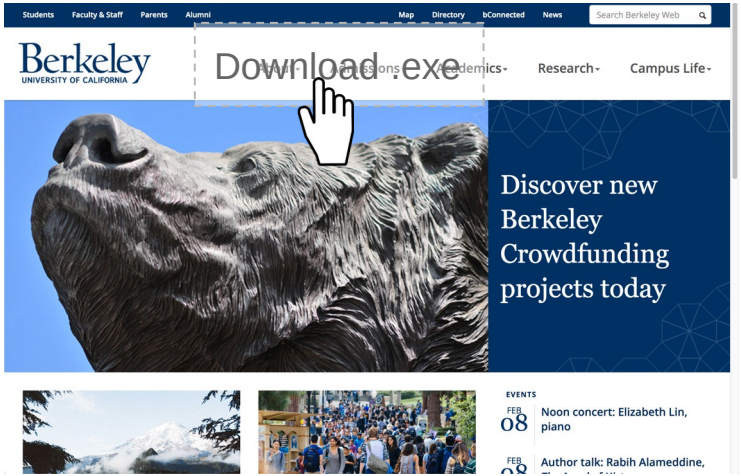
Clickjacking: Invisible iframes

- Variant #1: Frame the legitimate site invisibly, *over* visible, enticing content
  - Victim thinks they're clicking on the attacker's enticing website
  - But their click actually happened on the legitimate website!



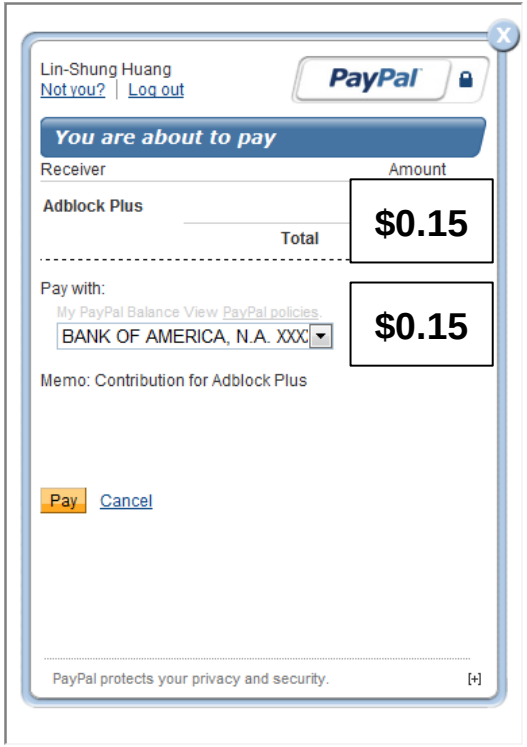
Clickjacking: Invisible iframes

- Variant #2: Frame the legitimate site visibly, *under* invisible malicious content
  - Victim thinks they're clicking on the legitimate site
  - But their click actually happened on the malicious website!



Clickjacking: Invisible iframes

- Variant #3: Frame the legitimate site visibly, *under* malicious content *partially* overlaying the site
  - The attacker can change the appearance of the site without breaking SOP!



Clickjacking: Temporal Attack

- JavaScript can detect the position of the cursor and change the website right before the user clicks on something
  - The user clicks on the malicious input (embedded iframe, download button, etc.) before they notice that something changed

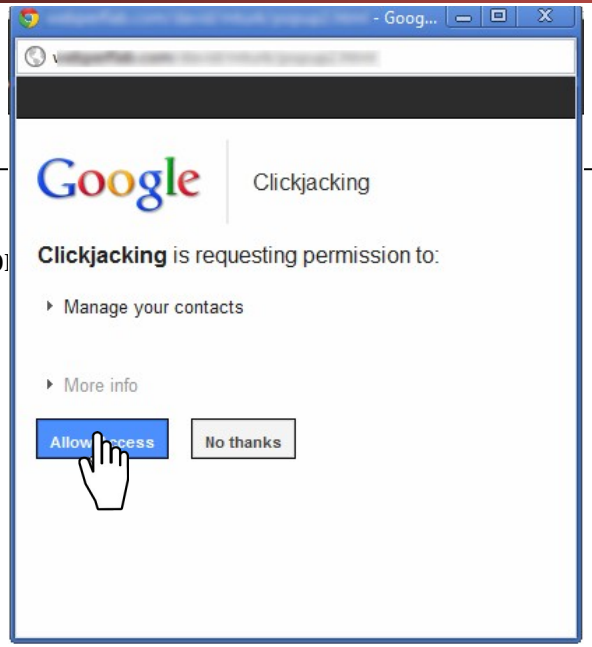
Clickjacking: Temporal Attack

Instructions:
Please double-click on the button below to continue to your content

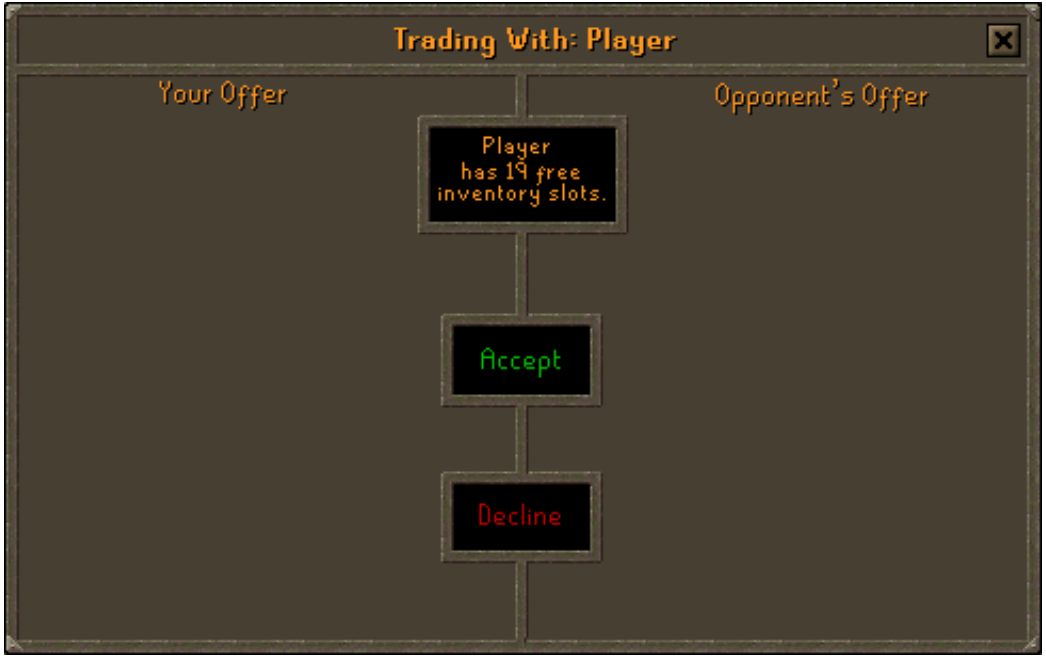
Click here

Clickjacking: Temporal Attack

Instructions:
Please double-click on the butto

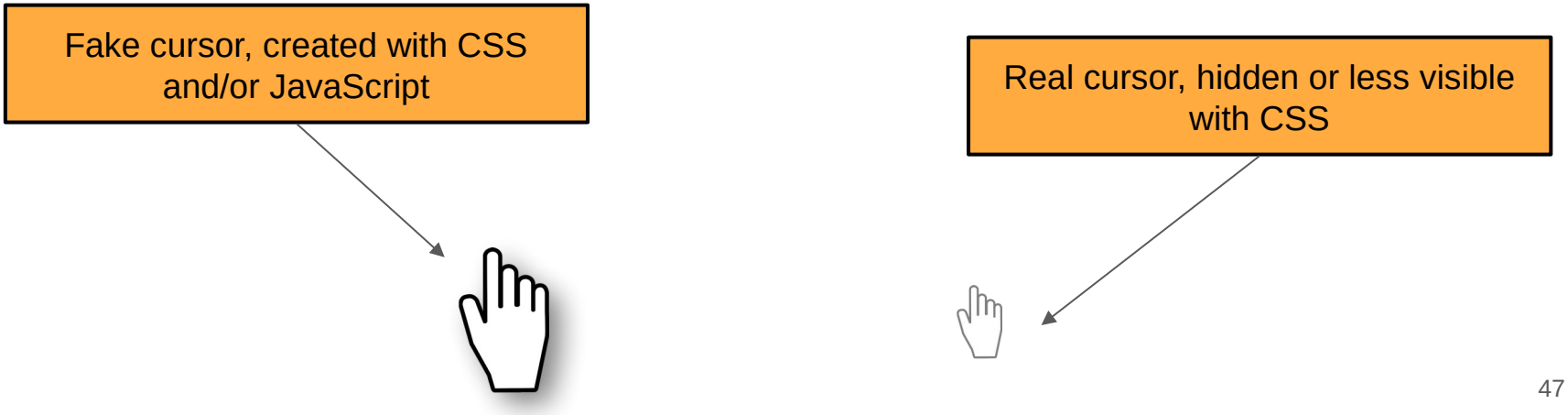


Clickjacking Example: Runescape Trade Scam



Clickjacking: Cursorjacking

- CSS has the ability to style the appearance of the cursor
- JavaScript has the ability to track a cursor's position
- If we change the appearance a certain way, we can create a fake cursor to trick users into clicking on things!



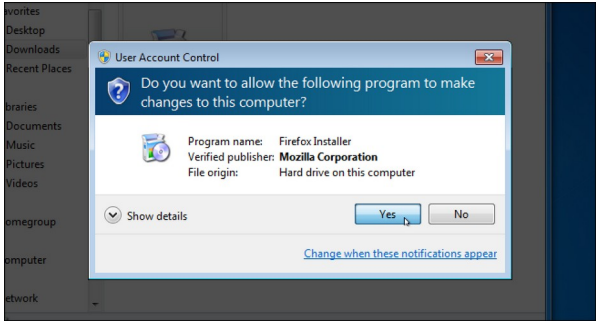
Clickjacking: Cursorjacking

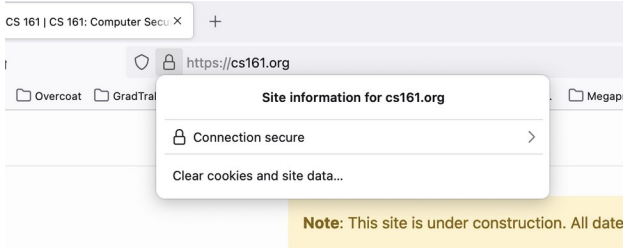
What do you think you're clicking on?

https://codesandbox.io/p/sandbox/ppr5gv

Clickjacking: Defenses

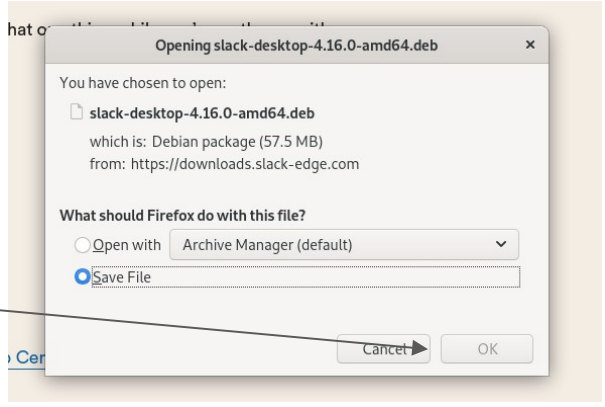
- **Enforce visual integrity**: Ensure clear visual separation between important dialogs and content
  - Windows User Account Control darkens the entire screen and freezes the desktop
  - Firefox dialogs “cross the boundary” between the URL bar and content, something that only valid dialogs can do

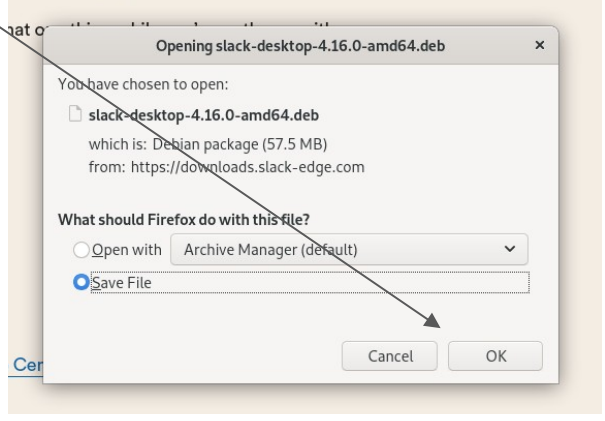




Clickjacking: Defenses

- **Enforce temporal integrity**: Ensure that there is sufficient time for a user to register what they are clicking on
  - Notice: Firefox blocks the “OK” button until 1 second after the dialog has been focused

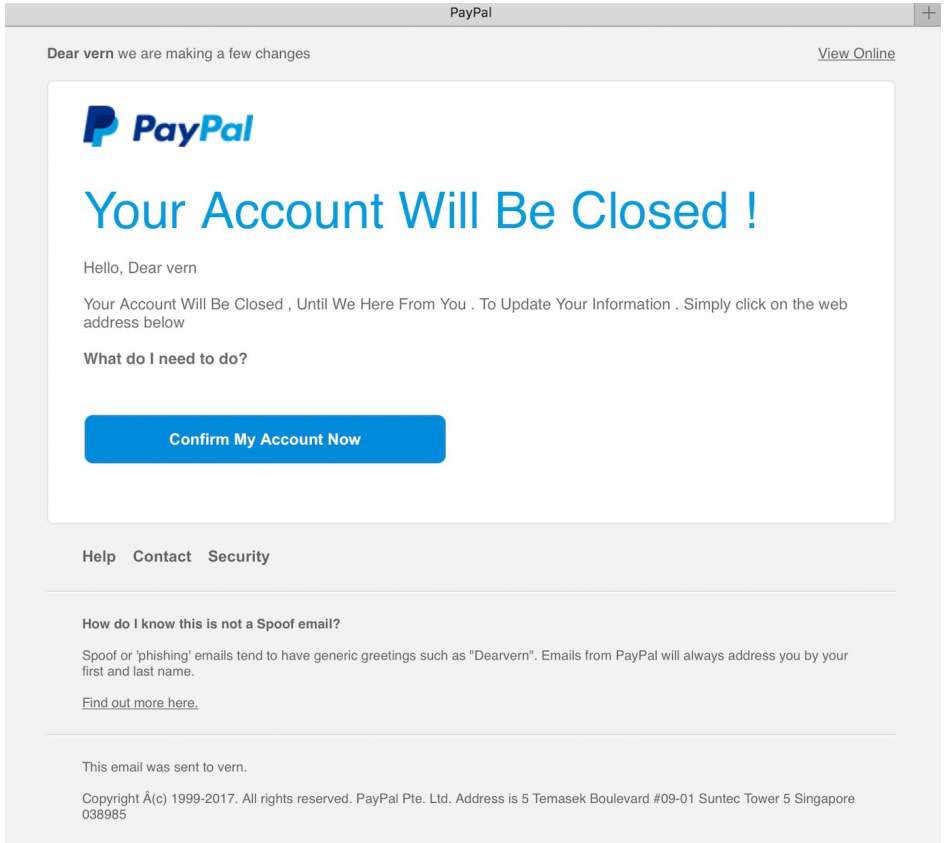




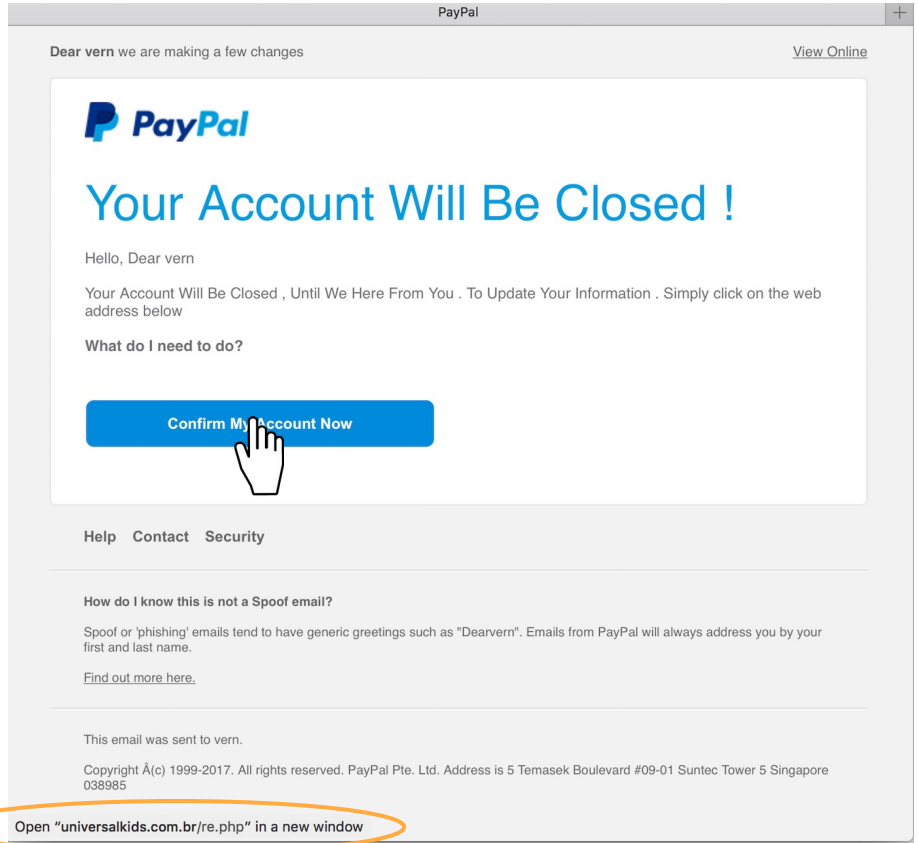
Clickjacking: Defenses

- **Require confirmation** from users
  - The browser needs to confirm that the user's click was intentional
  - Drawbacks: Asking for confirmation annoys users (consider human factors!)
- **Frame-busting**: The legitimate website forbids other websites from embedding it in an iframe
  - Defeats the invisible iframe attacks
  - Can be enforced by Content Security Policy (CSP)
  - Can be enforced by X-Frame-Options (an HTTP header)

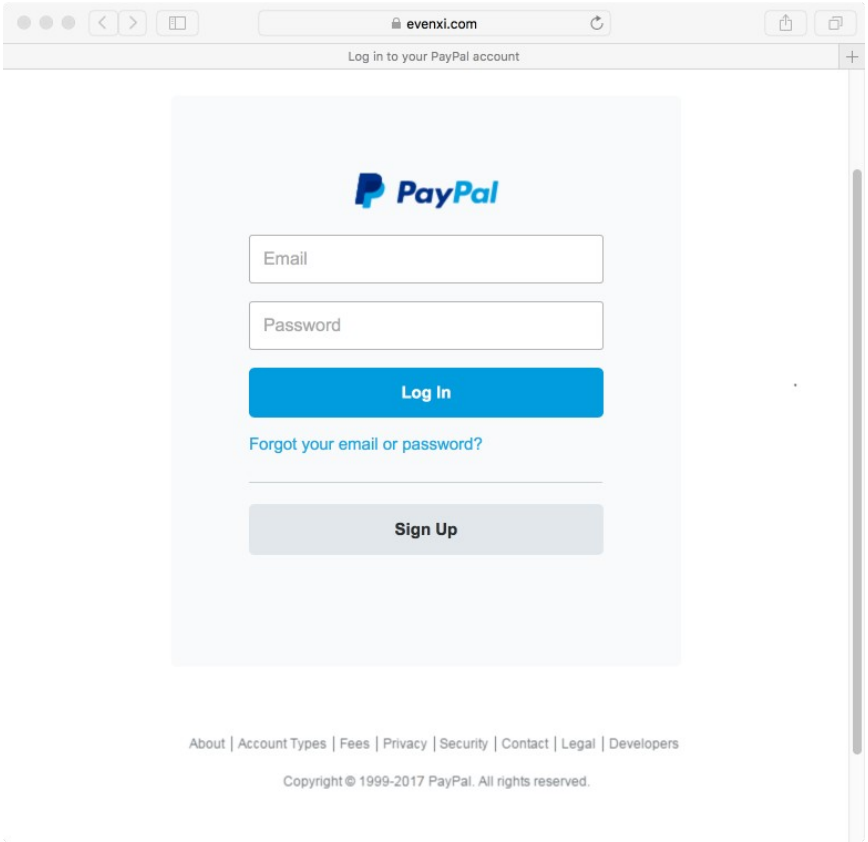
Phishing


PayPal
Dear vern we are making a few changes
View Online
PayPal
Your Account Will Be Closed !
Hello, Dear vern
Your Account Will Be Closed , Until We Here From You . To Update Your Information . Simply click on the web address below
What do I need to do?
Confirm My Account Now
Help Contact Security
How do I know this is not a Spoof email?
Spoof or 'phishing' emails tend to have generic greetings such as "Dearvern". Emails from PayPal will always address you by your first and last name.
Find out more here.
This email was sent to vern.
Copyright Â(c) 1999-2017. All rights reserved. PayPal Pte. Ltd. Address is 5 Temasek Boulevard #09-01 Suntec Tower 5 Singapore 038985

Phishing


PayPal
Dear vern we are making a few changes
View Online
PayPal
Your Account Will Be Closed !
Hello, Dear vern
Your Account Will Be Closed , Until We Here From You . To Update Your Information . Simply click on the web address below
What do I need to do?
Confirm My Account Now
Help Contact Security
How do I know this is not a Spoof email?
Spoof or 'phishing' emails tend to have generic greetings such as "Dearvern". Emails from PayPal will always address you by your first and last name.
Find out more here.
This email was sent to vern.
Copyright Â(c) 1999-2017. All rights reserved. PayPal Pte. Ltd. Address is 5 Temasek Boulevard #09-01 Suntec Tower 5 Singapore 038985
Open "universalkids.com.br/re.php" in a new window

Phishing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PayPal
Email
Password
Log In
Forgot your email or password?
Sign Up
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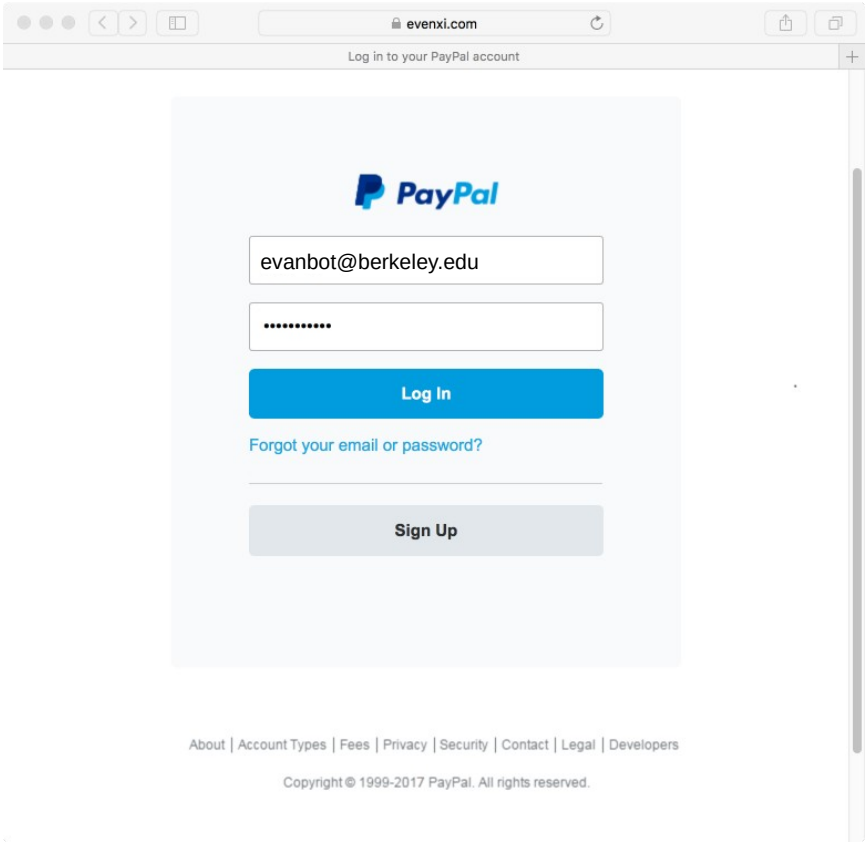
Phishing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PayPal
evanbot@berkeley.edu
Log In
Forgot your email or password?
Sign Up
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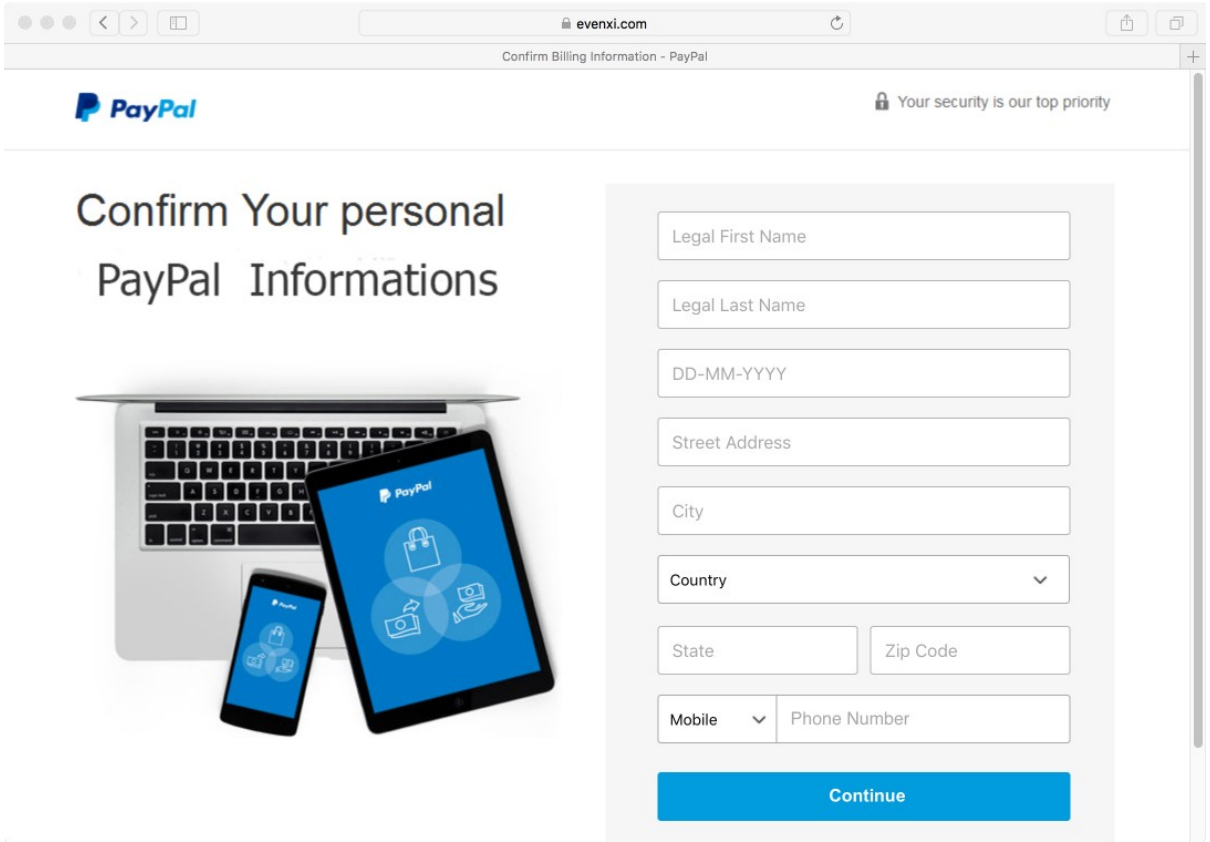
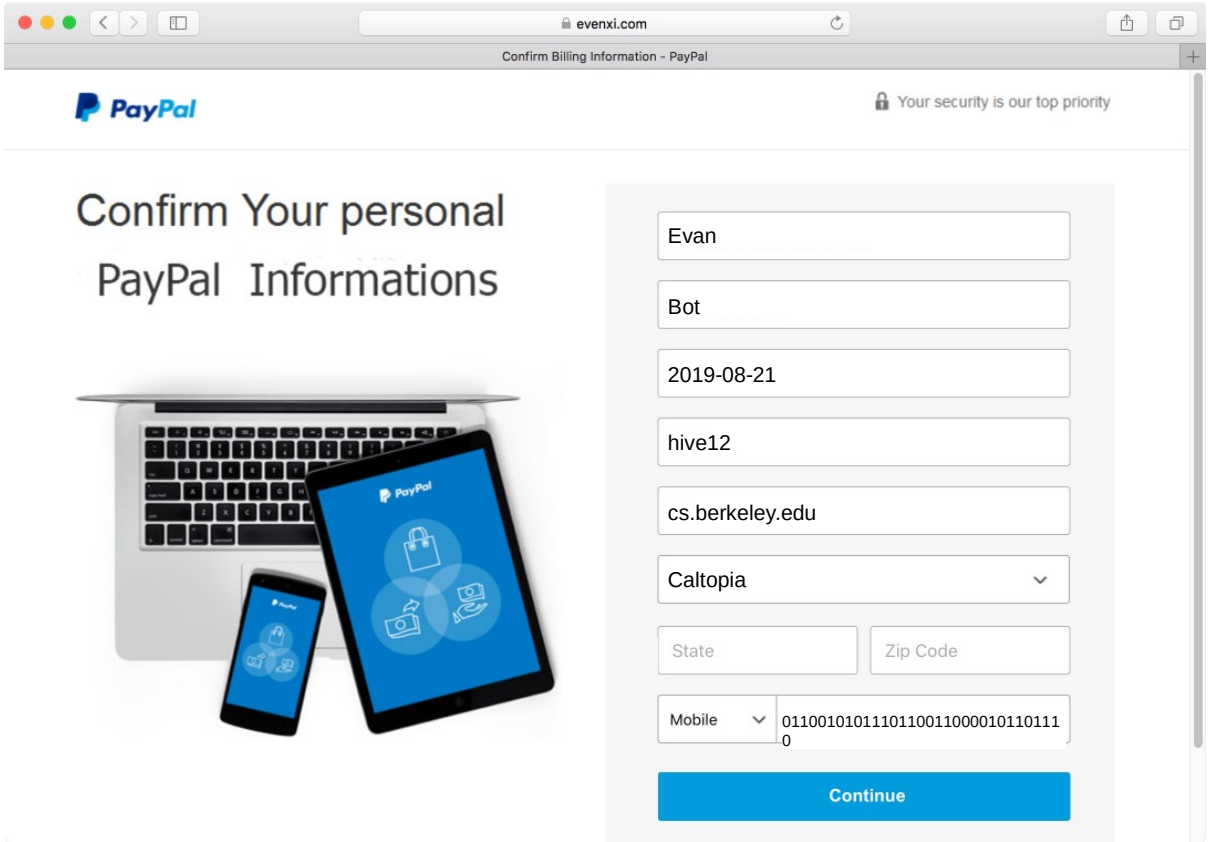
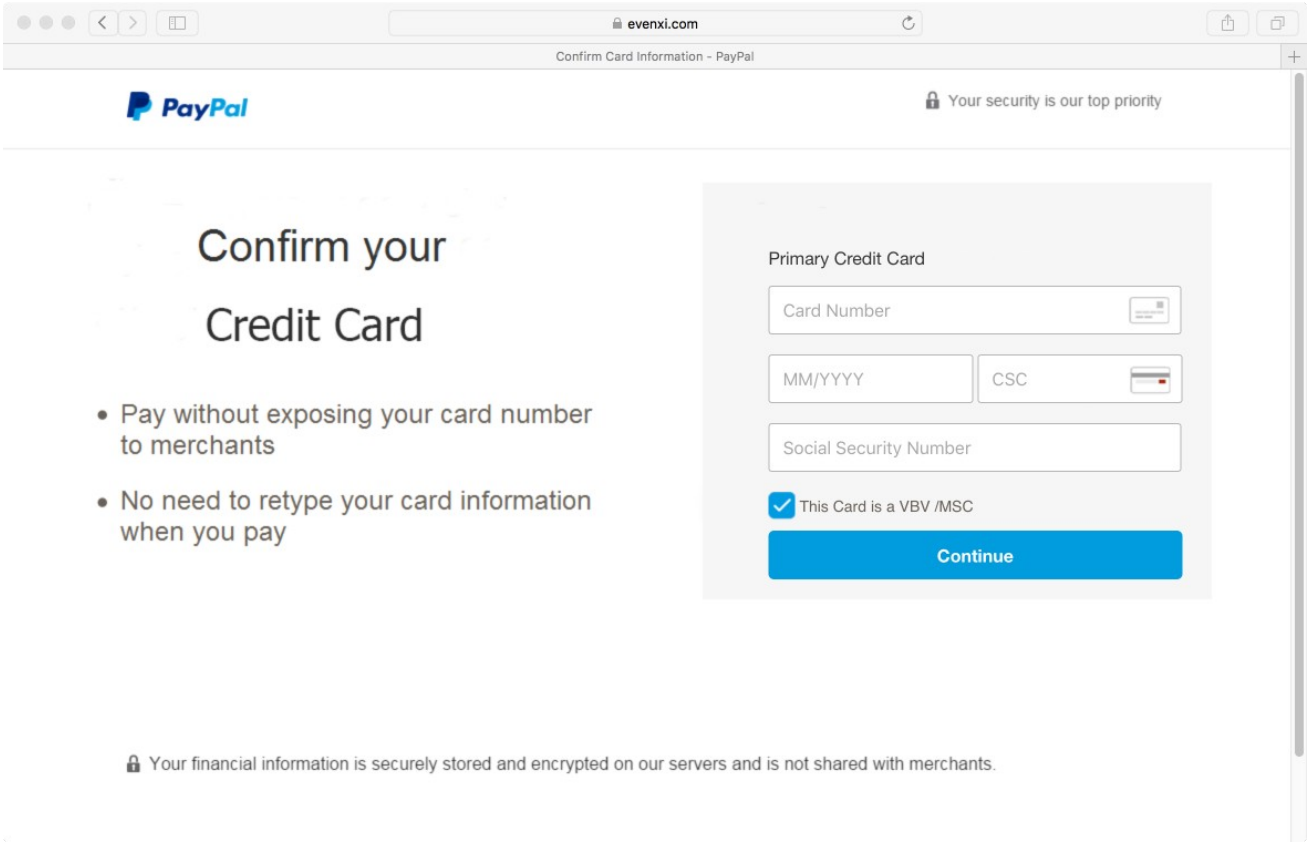
Phishing


evenxi.com
Confirm Billing Information - PayPal
PayPal
Your security is our top priority
Confirm Your personal
PayPal Informations
Legal First Name
Legal Last Name
DD-MM-YYYY
Street Address
City
Country
State
Zip Code
Mobile
Phone Number
Continue

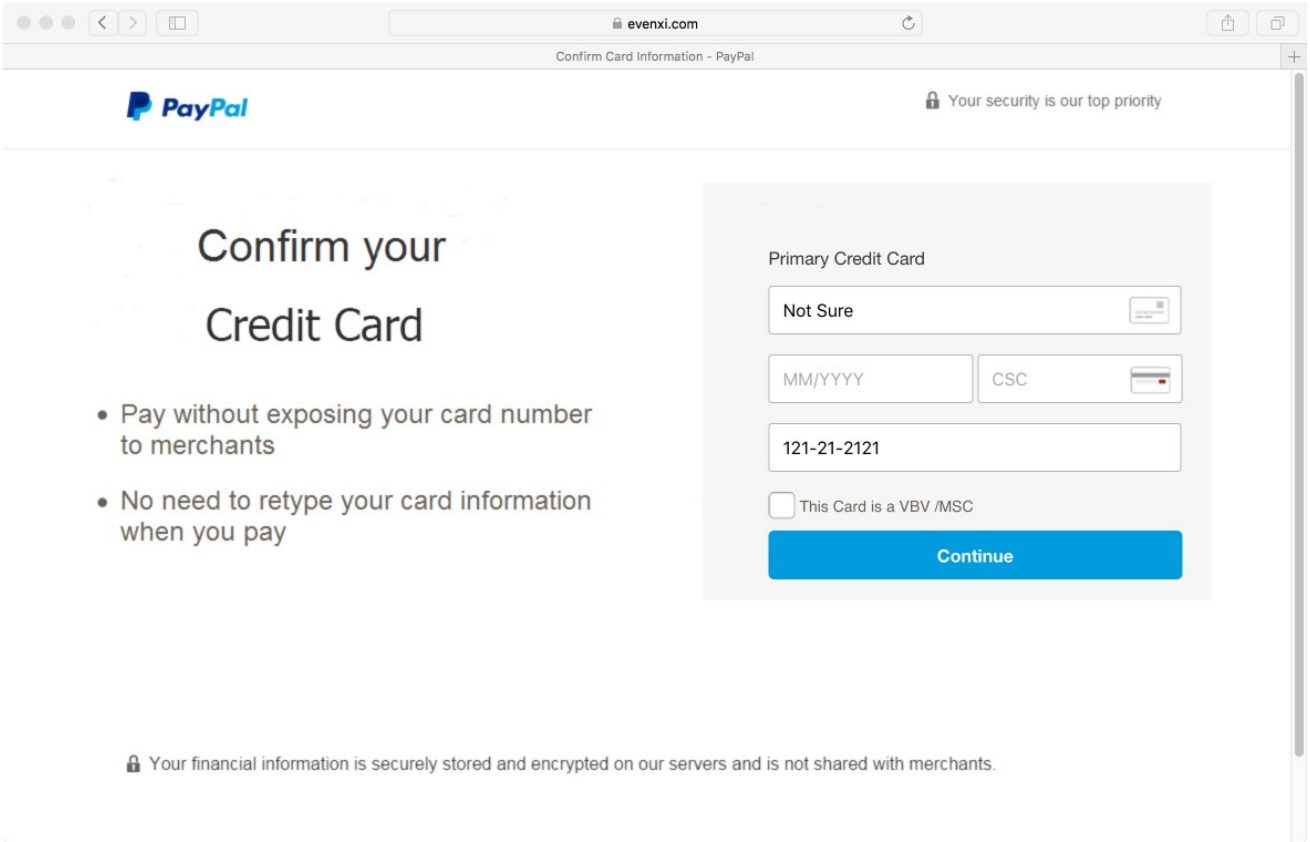
Phishing


evenxi.com
Confirm Billing Information - PayPal
PayPal
Your security is our top priority
Confirm Your personal
PayPal Informations
Evan
Bot
2019-08-21
hive12
cs.berkeley.edu
Caltopia
State
Zip Code
Mobile
01100101011101100110000101101110
Continue

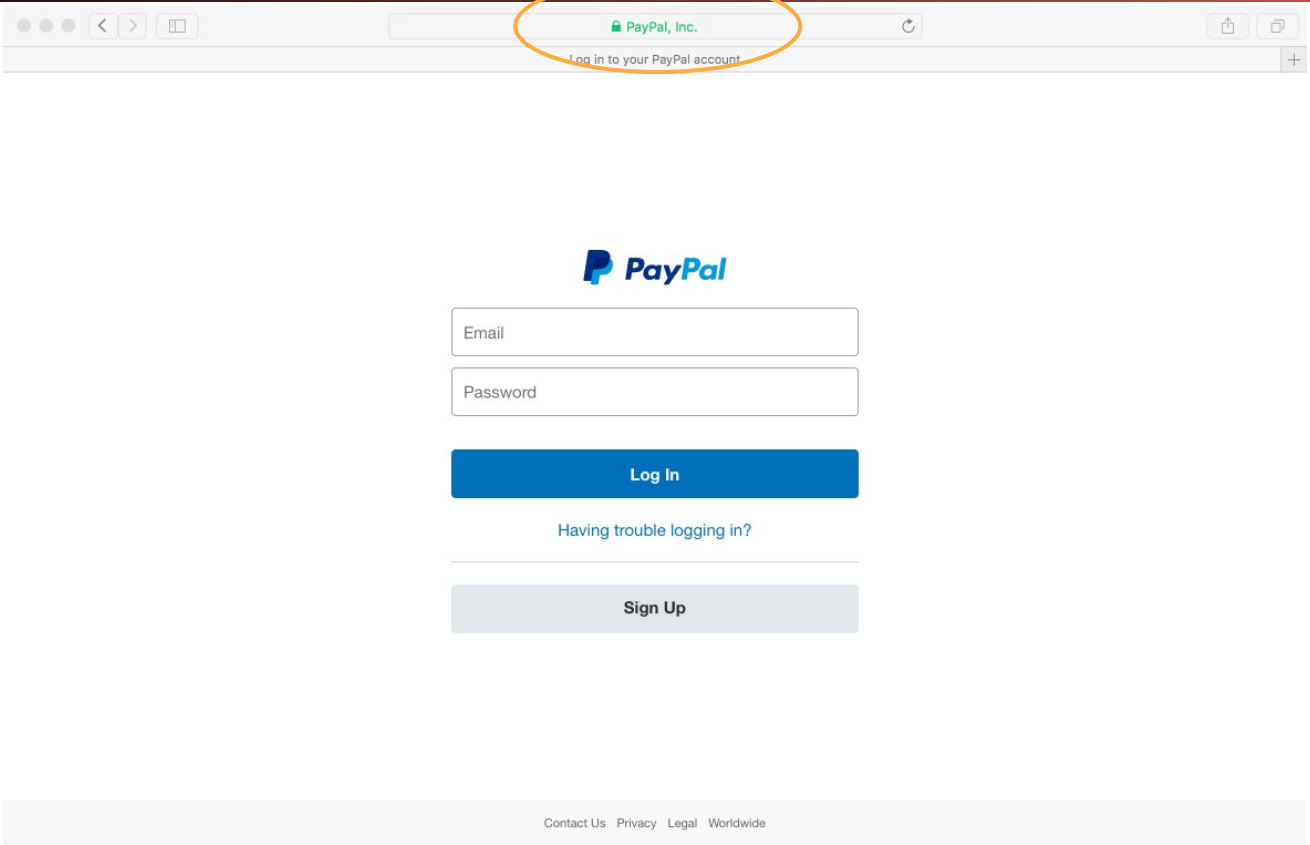
Phishing


evenxi.com
Confirm Card Information - PayPal
PayPal
Your security is our top priority
Confirm your
Credit Card
• Pay without exposing your card number to merchants
• No need to retype your card information when you pay
Primary Credit Card
Card Number
MM/YYYY
CSC
Social Security Number
This Card is a VBV /MSC
Continue
Your financial information is securely stored and encrypted on our servers and is not shared with merchants.

Phishing


evenxi.com
Confirm Card Information - PayPal
PayPal
Your security is our top priority
Confirm your
Credit Card
• Pay without exposing your card number to merchants
• No need to retype your card information when you pay
Primary Credit Card
Not Sure
MM/YYYY
CSC
121-21-2121
This Card is a VBV /MSC
Continue
Your financial information is securely stored and encrypted on our servers and is not shared with merchants.

Phishing

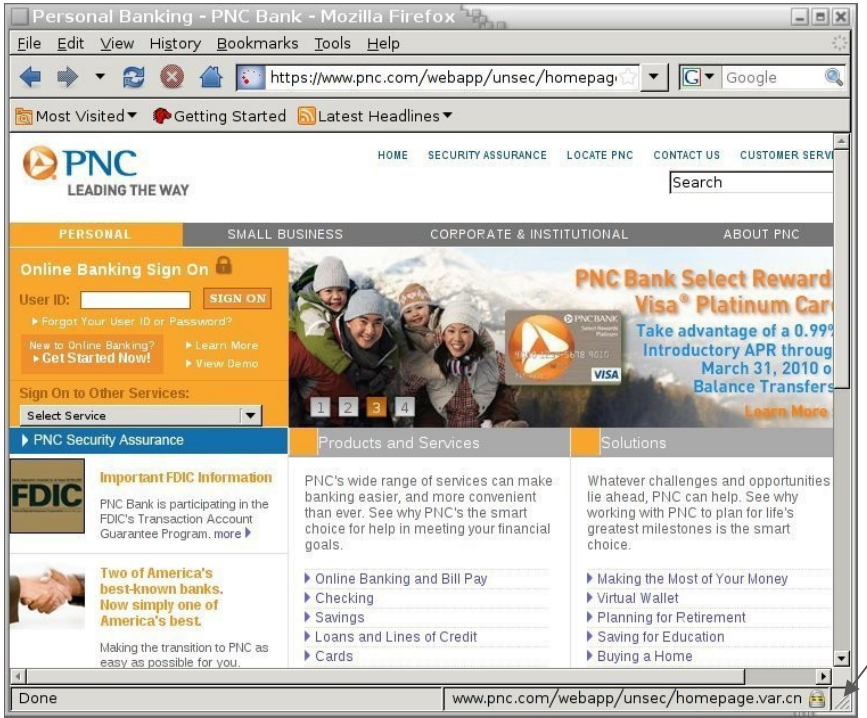

PayPal, Inc.
Log in to your PayPal account
PayPal
Email
Password
Log In
Having trouble logging in?
Sign Up
Contact Us Privacy Legal Worldwide

Phishing

- **Phishing**: Trick the victim into sending the attacker personal information
- Main vulnerability: The user can’t distinguish between a legitimate website and a website *impersonating* the legitimate website

Phishing: Check the URL?

Is this real?



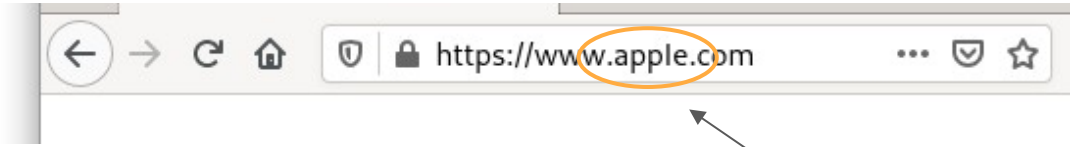
www.pnc.com/webapp/unsec/homepage.var.cn

~~is actually an entire domain!~~

The attacker can still register an HTTPS certificate for the perfectly valid domain

Phishing: Check the URL?

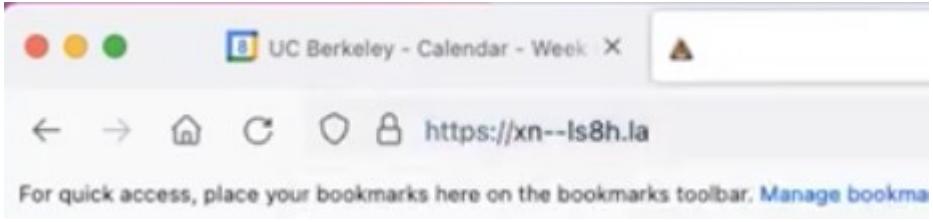
Is *this* real?



These letters come from the Cyrillic alphabet, not the Latin alphabet! They're rendered the same but have completely different bytes!

Phishing: Homograph Attacks

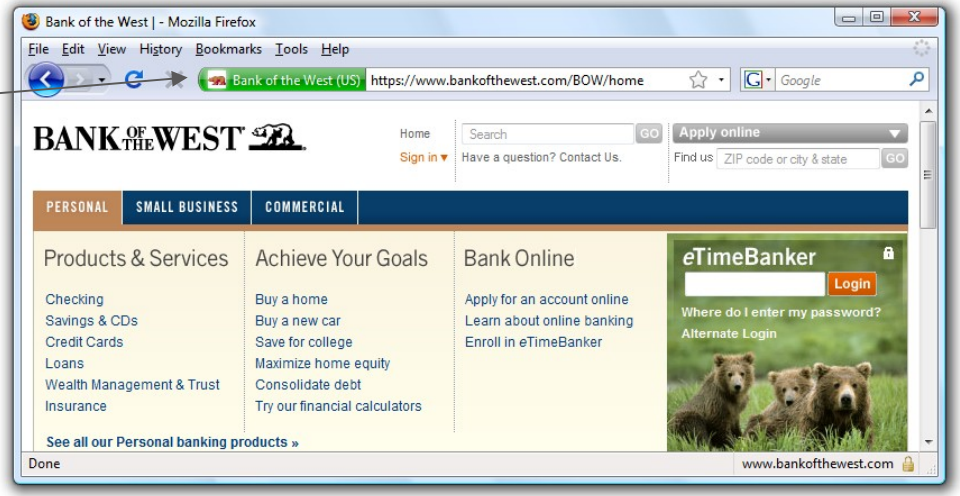
- Idea: Check if the URL is correct?
- **Homograph attack**: Creating malicious URLs that look similar (or the same) to legitimate URLs
  - Homograph: Two words that look the same, but have different meanings



Phishing: Check *Everything*

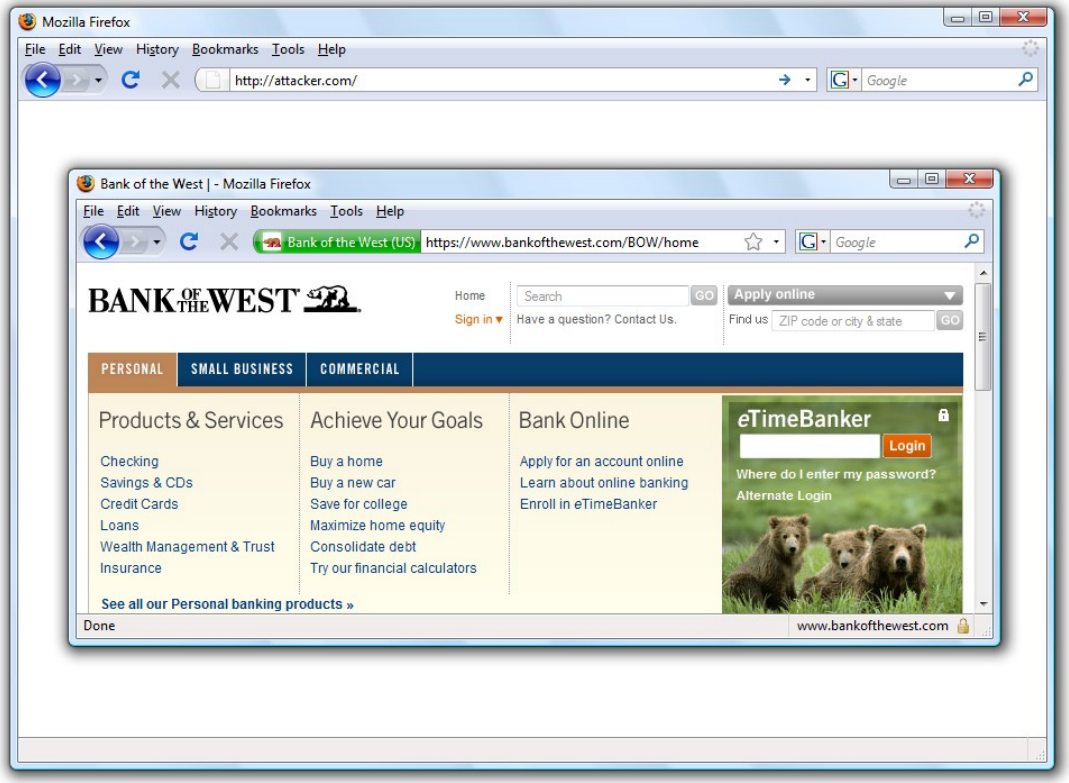
Is *this* real?

Extended Validation: Certificate authority verified the identity of the site (not just the domain)



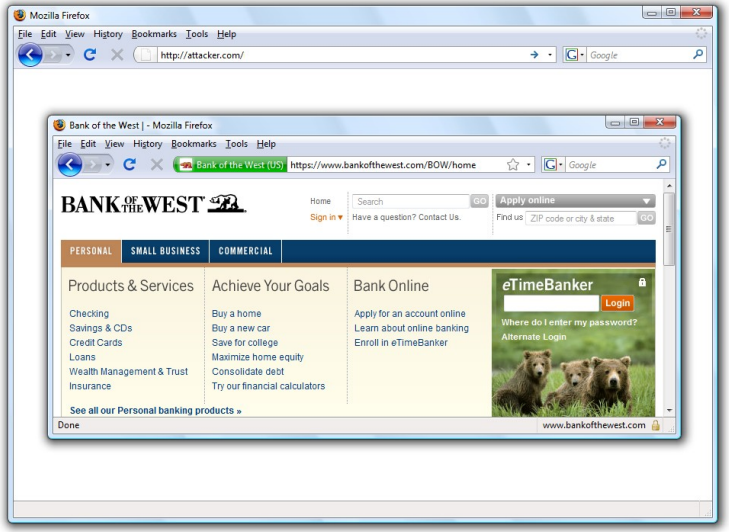
Phishing: Check *Everything*

Oops, never mind



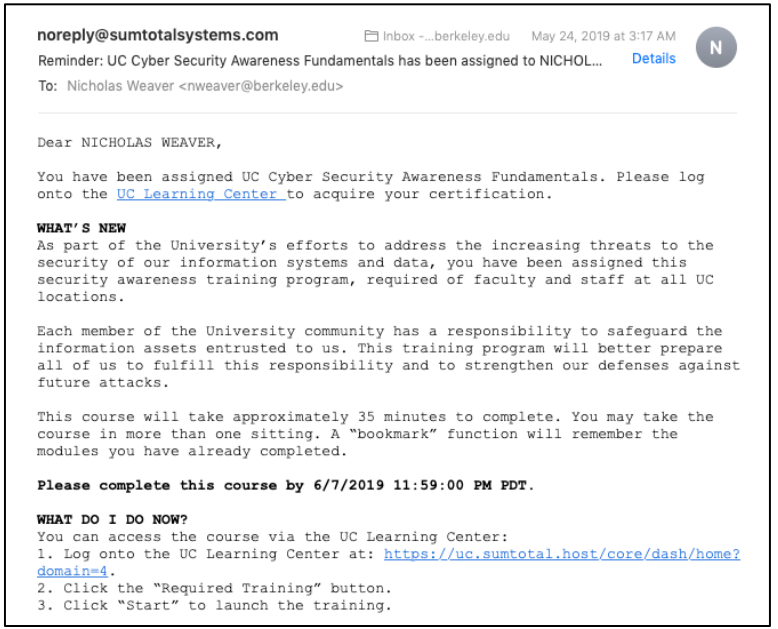
Phishing: Browser-in-browser Attacks

- Idea: Check for a green padlock icon in the browser's address bar, or any other built-in browser security feature
- **Browser-in-browser attack**: The attacker simulates the entire web browser with JavaScript



Phishing: Don’t Blame the Users

- Most users aren’t security experts
- Attacks are uncommon: users normally don't suspect malicious action
- Detecting phishing is hard, even if you’re on the lookout for attacks
  - Legitimate messages often look like phishing attacks!



noreply@sumtotalsystems.com Inbox -...berkeley.edu May 24, 2019 at 3:17 AM

Reminder: UC Cyber Security Awareness Fundamentals has been assigned to NICHOL... Details

To: Nicholas Weaver <nweaver@berkeley.edu>

Dear NICHOLAS WEAVER,

You have been assigned UC Cyber Security Awareness Fundamentals. Please log onto the UC Learning Center to acquire your certification.

WHAT'S NEW
As part of the University's efforts to address the increasing threats to the security of our information systems and data, you have been assigned this security awareness training program, required of faculty and staff at all UC locations.

Each member of the University community has a responsibility to safeguard the information assets entrusted to us. This training program will better prepare all of us to fulfill this responsibility and to strengthen our defenses against future attacks.

This course will take approximately 35 minutes to complete. You may take the course in more than one sitting. A "bookmark" function will remember the modules you have already completed.

Please complete this course by 6/7/2019 11:59:00 PM PDT.

WHAT DO I DO NOW?
You can access the course via the UC Learning Center:
1. Log onto the UC Learning Center at: https://uc.sumtotal.host/core/dash/home?domain=4.
2. Click the "Required Training" button.
3. Click "Start" to launch the training.

Two-Factor Authentication

- Problem: Phishing attacks allow attackers to learn passwords
- Idea: Require more than passwords to log in
- **Two-factor authentication** (**2FA**): The user must prove their identity in two different ways before successfully authenticating
- Three main ways for a user to prove their identity
  - **Something the user knows**: Password, security question (e.g. name of your first pet)
  - **Something the user has**: Their phone, their security key
  - **Something the user is**: Fingerprint, face ID
- Even if the attacker steals the user's password with phishing, they don't have the second factor

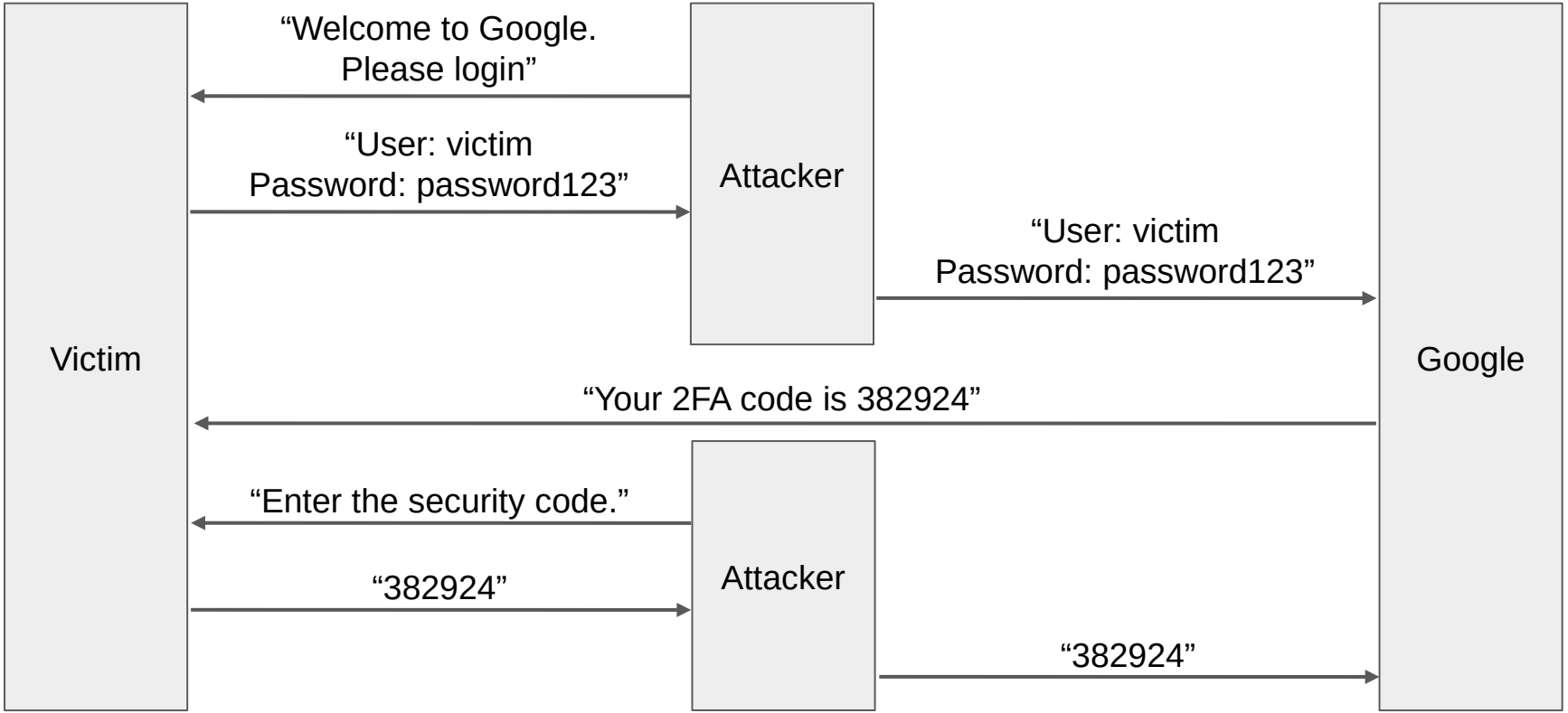
Two-Factor Authentication

- Two-factor authentication also defends against other attacks where a user's password is compromised
  - Example: An attacker steals the password file and performs a dictionary attack
  - Example: The user reuses passwords on two different websites. The attacker compromises one website and tries the same password on the second website
  - With 2FA, the password alone is no longer enough for the attacker to log in

Subverting 2FA: Relay Attacks

- **Relay attacks** (**transient phishing**): The attacker steals both factors in a phishing attack
- Example
  - Two-factor authentication scheme
    - First factor: The user's password (something the user knows)
    - Second factor: A code sent to the user's phone (something the user owns)
  - Attack
    - The phishing website asks the user to input their password (first factor)
    - The attacker immediately tries to log in to the actual website as the user
    - The actual website sends a code to the user
    - The phishing website asks the user to enter the code (second factor)
    - The attacker enters the code to log in as the user

Subverting 2FA: Relay Attacks



Subverting 2FA: Social Engineering

- Some 2FA schemes text a one-time code to a phone number
  - Attackers can call your phone provider (e.g., Verizon) and tell them to activate the attacker's SIM card, so they receive your texts!
  - 2FA via SMS is not great but *better than nothing*
- Some 2FA schemes can be bypassed with customer support
  - Attackers can call customer support and ask them to deactivate 2FA!
  - Companies should validate identity if you ask to do this (but not all do)

Summary: XSS

- Occurs when websites use untrusted content as control data
  - `<html><body>Hello EvanBot!</body></html>`
  - `<html><body>Hello <script>alert(1)</script>!</body></html>`
- Stored XSS
  - The attacker's JavaScript is stored on the legitimate server and sent to browsers
  - Classic example: Make a post on a social media site (e.g. Facebook) with JavaScript
- Reflected XSS
  - The attacker causes the victim to input JavaScript into a request, and the content it's reflected (copied) in the response from the server
  - Classic example: Create a link for a search engine (e.g. Google) query with JavaScript
  - Requires the victim to click on the link with JavaScript

Summary: XSS Defenses

- Defense: HTML sanitization
  - Replace control characters with data sequences
    - **<** becomes **`<`**
    - **"** becomes **`&quot;`**
  - Use a trusted library to sanitize inputs for you
- Defense: Content Security Policy (CSP)
  - Instruct the browser to only use resources loaded from specific places
  - Limits JavaScript: only scripts from trusted sources are run in the browser
  - Enforced by the browser

Summary: Clickjacking

- Clickjacking: Trick the victim into clicking on something from the attacker
- Main vulnerability: the browser trusts the user's clicks
  - When the user clicks on something, the browser assumes the user intended to click there
- Examples
  - Fake download buttons
  - Show the user one frame, when they're actually clicking on another invisible frame
  - Temporal attack: Change the cursor just before the user clicks
  - Cursorjacking: Create a fake mouse cursor with JavaScript
- Defenses
  - Enforce visual integrity: Focus the user's vision on the relevant part of the screen
  - Enforce temporal integrity: Give the user time to understand what they're clicking on
  - Ask the user for confirmation
  - Frame-busting: The legitimate website forbids other websites from embedding it in an iframe

Summary: Phishing

- Phishing: Trick the victim into sending the attacker personal information
  - A malicious website impersonates a legitimate website to trick the user
- Don’t blame the users
  - Detecting phishing is hard, especially if you aren’t a security expert
  - Check the URL? Still vulnerable to homograph attacks (malicious URLs that look legitimate)
  - Check the entire browser? Still vulnerable to browser-in-browser attacks
- Defense: Two-Factor Authentication (2FA)
  - User must prove their identity two different ways (something you know, something you own, something you are)
  - Defends against attacks where an attacker has only stolen one factor (e.g. the password)
  - Vulnerable to relay attacks: The attacker phishes the victim into giving up both factors
  - Vulnerable to social engineering attacks: Trick humans to subvert 2FA
  - Example: Authentication tokens for generating secure two-factor codes
  - Example: Security keys to prevent phishing